PROJECT DOCUMENTATION

Script-Controlled ACL – Restrict Record Access Based on Field Value

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Team ID	SWTID-2026-9980
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

1. INTRODUCTION

1.1 Project Overview

This project focuses on implementing field-value-based access control using the ServiceNow platform. In enterprise applications, not every user should be allowed to view or modify every record. Access may depend on the value stored in a specific field, such as department, access level, confidentiality, ownership, or record status.

The solution uses ServiceNow Access Control Lists (ACLs), roles, table/field permissions, and server-side JavaScript scripting to evaluate record values before granting access. The script-controlled ACL dynamically allows or denies record access according to the defined field-value condition.

1.2 Purpose

The purpose of this project is to build a secure and maintainable ServiceNow access-control mechanism that restricts record access based on field values. Key goals include:

- Creating an ACL for the required table and operation such as Read or Write.
- Checking the value of a selected record field inside the ACL script.
- Granting access only when the record satisfies the defined field-value condition and the user has the required role or permission.
- Preventing unauthorized users from accessing restricted records while keeping the rule manageable through server-side scripting.

2. IDEATION PHASE

2.1 Problem Statement

Enterprise ServiceNow tables may contain records with different levels of sensitivity or ownership. A simple role-only permission may allow users to access records that should be restricted according to their field values.

- Field-based restriction gap: Users with the same role may still need different access depending on a record's field value.
- Unauthorized record access: Without a scripted condition, restricted records may be visible or editable to users who should not access them.
- Difficult maintenance: Multiple hard-coded restrictions can become difficult to manage without a clear ACL rule and documented condition.

2.2 Empathy Map Canvas

- Says: "Users should only access records that they are permitted to see based on the record details."
- Thinks: "The system should automatically check the field value instead of depending on manual verification."
- Does: Attempts to open, read, or modify a record through the ServiceNow interface according to the assigned role and access level.

- Pains: Unclear access rules, accidental exposure of restricted records, and repeated manual permission checks.
- Gains: Consistent access decisions, reduced unauthorized access, centralized security logic, and easier maintenance.

2.3 Brainstorming

- Option A: Role-only ACL – Simple to configure, but it cannot distinguish records based on their field values.
- Option B: Client-side visibility logic – Can improve the interface experience, but it should not be treated as the primary server-side security control.
- Option C: Script-Controlled ACL based on field value – Evaluates the record field at the ACL level and grants or denies access according to the defined condition.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- Identify: Administrator identifies the table, field, operation, role, and restricted value that require protection.
- Configure: Administrator creates the appropriate ServiceNow ACL and defines the required roles and condition.
- Script: The ACL script checks the target record's field value and sets the access result accordingly.
- Validate: Authorized and unauthorized users attempt to access the record to verify that the rule behaves correctly.
- Maintain: Administrator reviews the ACL logic and updates the field-value rule when the application security requirements change.

3.2 Solution Requirement

Functional Requirements (FR)

- FR-1: Create a ServiceNow ACL for the selected table and operation, such as Read or Write.
- FR-2: Identify and evaluate the required record field within the ACL script.
- FR-3: Grant access when the configured field-value condition and required role/permission are satisfied.
- FR-4: Deny access when the record contains a restricted field value or the user does not satisfy the required access rule.

Non-Functional Requirements (NFR)

- NFR-1 (Security): Access decisions must be enforced at the ServiceNow ACL/server side.
- NFR-2 (Maintainability): The ACL script should be clear, concise, and easy to update.
- NFR-3 (Reliability): The rule should consistently produce the same access decision for the same inputs.
- NFR-4 (Performance): The ACL should perform only the required field-value and permission checks.
- NFR-5 (Auditability): ACL configuration and access-control logic should be documented for review.
- NFR-6 (Scalability): The approach should be reusable for additional field-value restrictions and tables.

3.3 Data Flow Diagram

USER / REQUEST → SERVICENOW TABLE RECORD → ACL EVALUATION → ROLE/PERMISSION CHECK + FIELD-VALUE CHECK → ACCESS DECISION → ALLOW / DENY

3.4 Technology Stack

- User Interface: ServiceNow forms and list views.
- Security Configuration: ACL records, table/field permissions, roles, and conditions.

- Server-Side Logic: ACL Script using ServiceNow server-side JavaScript.
- Platform: ServiceNow Cloud Instance / Personal Developer Instance (PDI).
- Data: Protected ServiceNow table and the field used for access evaluation.
- Validation: Test users, roles, restricted records, and positive/negative test cases.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution addresses field-based record security by moving the access decision into a ServiceNow ACL. Instead of relying only on the user's role, the ACL evaluates the actual record field value before allowing the requested operation.

4.2 Proposed Solution

Implement a Script-Controlled ACL in ServiceNow for the selected table and operation. The ACL checks the user's required permission and reads the configured field from the current record. If the field value matches the permitted condition, the script grants access; otherwise, access is denied.

4.3 Solution Architecture

User → Record Request → ACL Evaluation → Role/Permission Check + Field-Value Script Check → Access Decision → Allow or Deny

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Sprint 1: Identify the protected table, target field, operation, roles, permitted value, and restricted value. Create the ACL record.
- Sprint 2: Implement the ACL script, configure the rule, test authorized and unauthorized access, and document the final security behaviour.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Testing

- Test Case 1 – Authorized Value: A user satisfying the required role/permission accesses a record with the permitted field value. Expected: access allowed.
- Test Case 2 – Restricted Value: A user attempts to access a record whose field contains the restricted value. Expected: access denied.
- Test Case 3 – Different Record Value: The same user accesses a record with a different field value to verify that the ACL decision changes according to the rule.
- Test Case 4 – Administrator Access: An administrator with the appropriate permission accesses the protected record as configured by the ACL.

7. RESULTS

7.1 Output Screenshots

The following screenshots illustrate the project configuration, script logic, record-level field value, denied access behaviour, roles, and validation results.

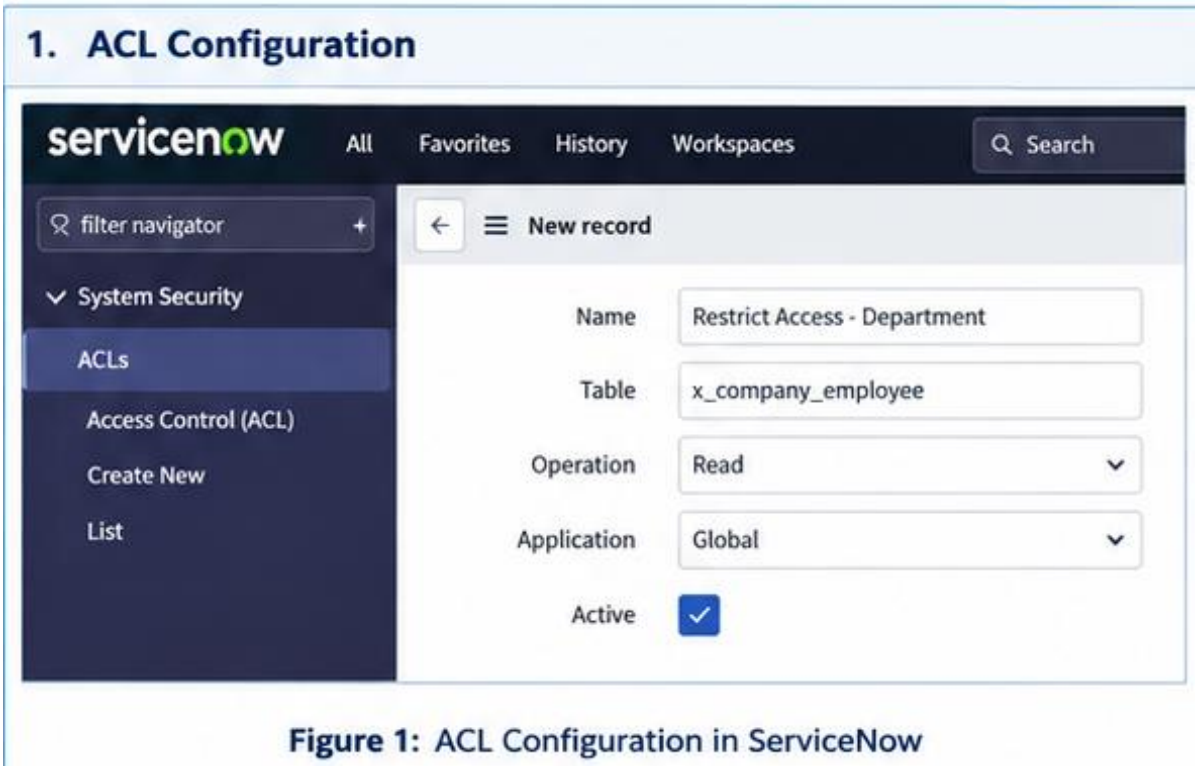


Figure 1: ACL Configuration in ServiceNow

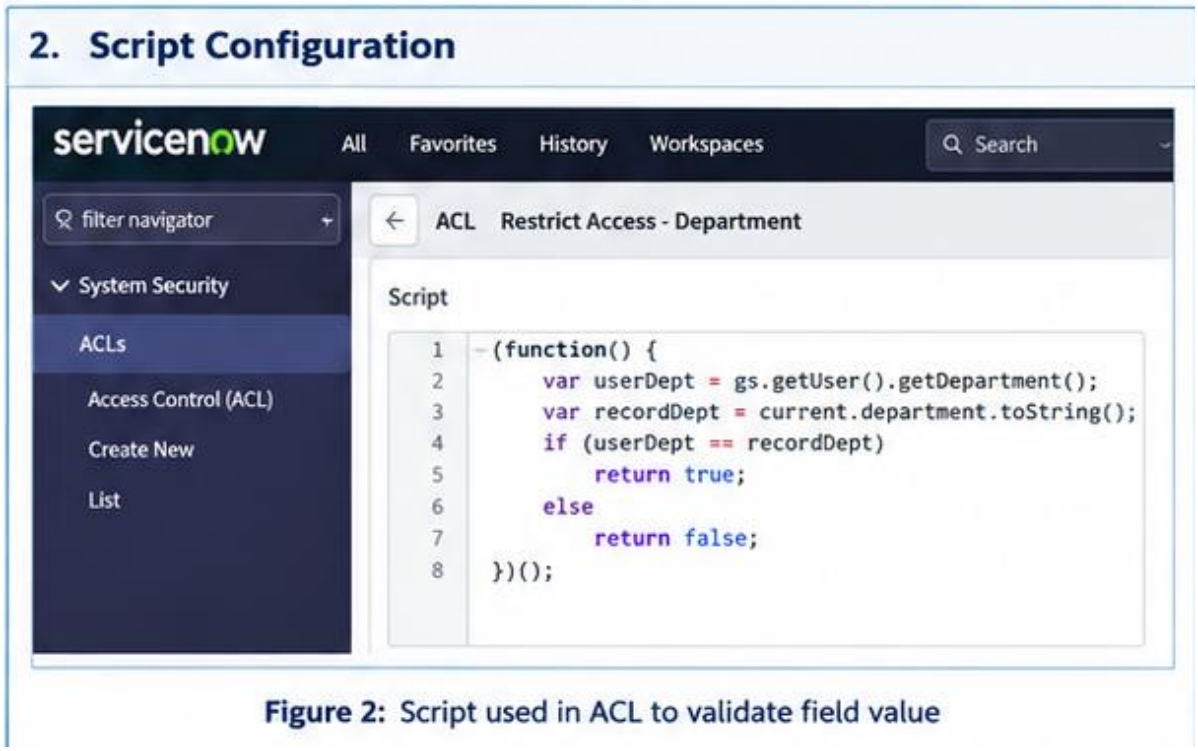


Figure 2: Script used in ACL to validate field value

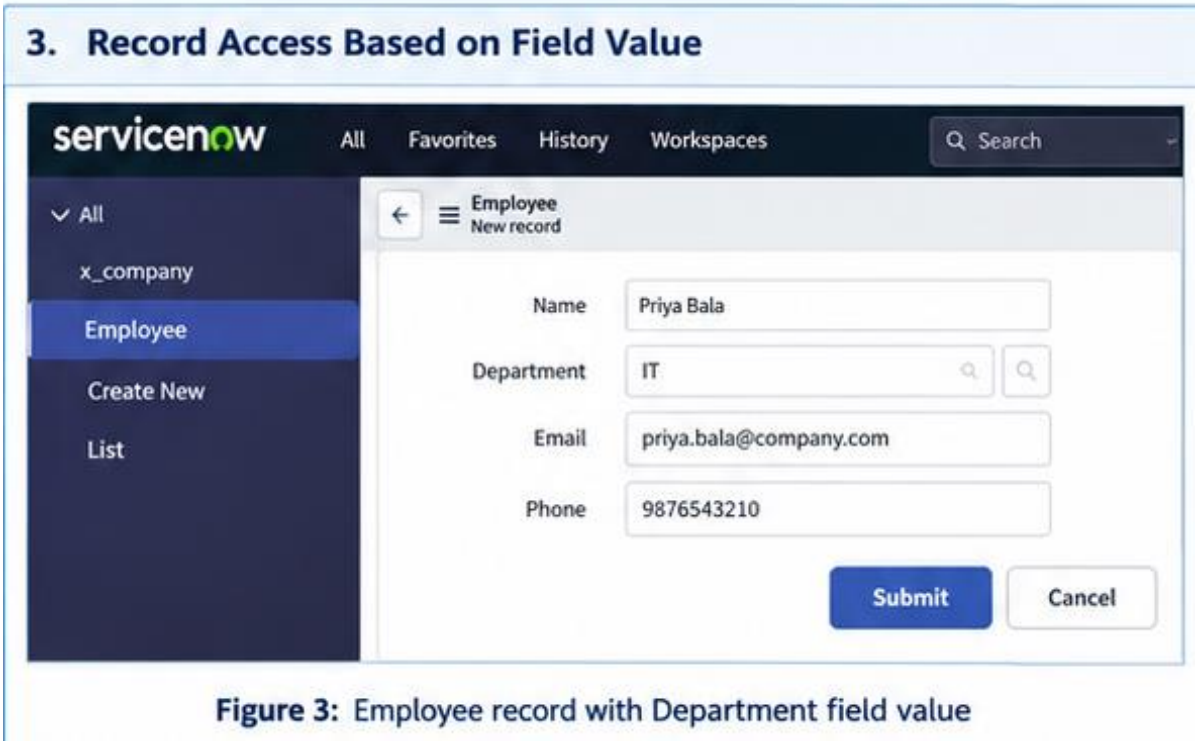


Figure 3: Employee record with Department field value

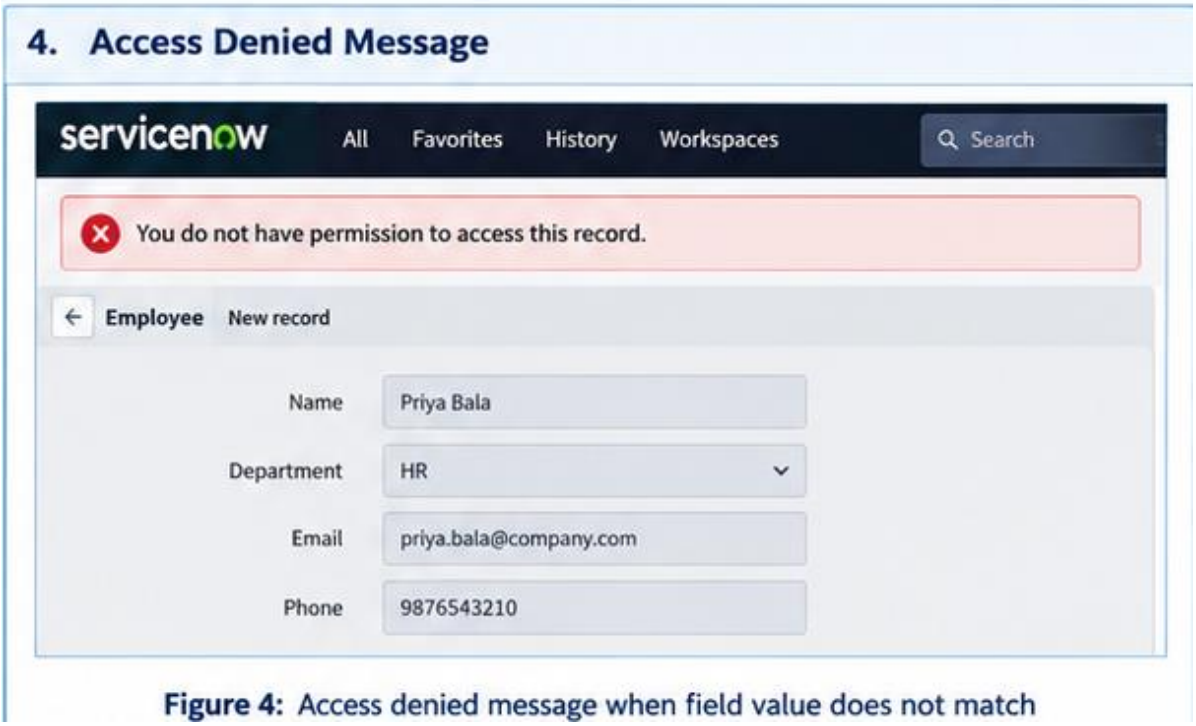


Figure 4: Access denied when field value does not match

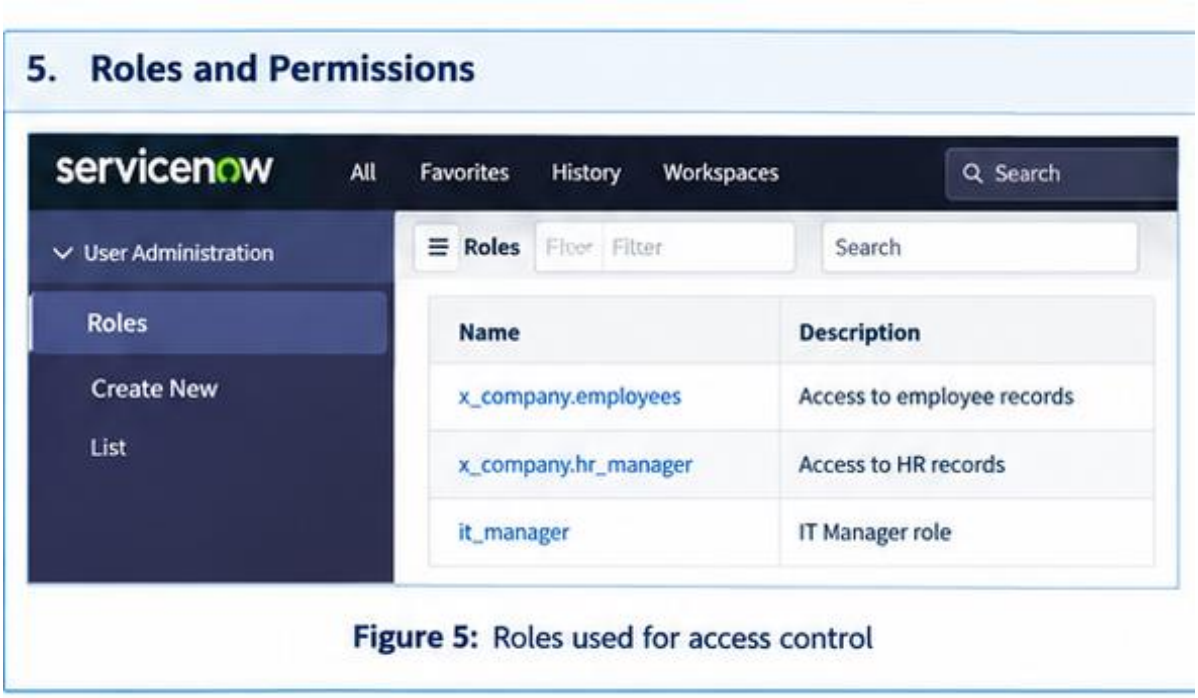


Figure 5: Roles used for access control

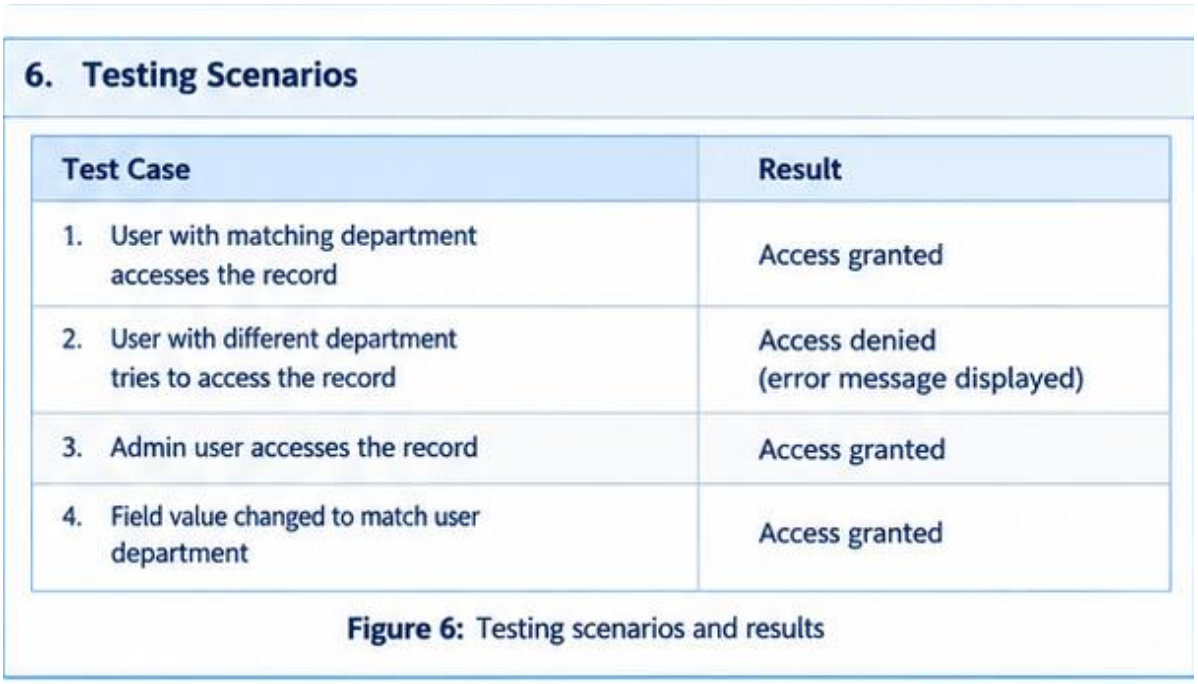


Figure 6: Testing scenarios and results

7.2 Result Summary

The documented configuration demonstrates that the ACL evaluates the record field value and produces an access decision. Matching conditions result in access being granted, while non-matching restricted conditions result in access being denied.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Field-Based Security: Access decisions can depend on the actual value stored in a record.
- Centralized Control: Security logic is maintained in ServiceNow ACL configuration.
- Reusability: Similar ACL logic can be adapted for other tables, fields, and operations.
- Reduced Unauthorized Access: Restricted records can be denied at the server-side ACL layer.

8.2 Disadvantages

- Script Complexity: Poorly written ACL scripts can make security rules difficult to maintain.
- Testing Requirement: Multiple roles, operations, and field values must be tested.
- Configuration Dependency: Changes to field names or table structure may require updates to the ACL script.

9. CONCLUSION

The Script-Controlled ACL – Restrict Record Access Based on Field Value solution provides a practical ServiceNow security mechanism for controlling record access dynamically. By combining ACL configuration, user permissions, and server-side field-value validation, the project ensures that access decisions are made according to the defined security rule. The approach is maintainable, reusable, and suitable for demonstrating secure record-level access control in ServiceNow.